

Appendix 2: Aptitude Assessment

Aptitude Assessment for the Master's Degree Program in Geodesy and Geoinformation at the Technical University of Munich

1. Purpose of the Process

¹Eligibility for the Master's Degree Program in Geodesy and Geoinformation, in addition to the requirements according to § 36(1)1 and 2, requires proof of aptitude as set out in § 36(1)3 in accordance with the following provisions. ²The special qualifications and skills of the candidates should correspond to the field of geodesy. ³Individual aptitude parameters are:

- 1.1 ability to conduct scholarly and/or basic and methodologically sound research;
- 1.2 specialist knowledge in the field of geodesy from a bachelor's degree program comparable to the Bachelor's Degree Program in Geodesy and Geoinformation at the Technical University of Munich,
- 1.3 a sound knowledge of the fundamental sciences of geodesy, i.e. fundamental knowledge of mathematics, informatics, statistics and physics,
- 1.4 an affinity for engineering and interest in geodetic questions, as well as the ability to think in spatial terms,
- 1.5 ability to work interdisciplinarily, and command of the English language.

2. Aptitude Assessment Process

- 2.1 Aptitude Assessment takes place annually (semi-annually for applicants in higher semesters).
- 2.2 ¹Applications for admission to the aptitude assessment process for the winter semester must be submitted to the Technical University of Munich together with the documents listed in 2.3.1. through 2.3.4. and in § 36(1)2 no later than 31 May (absolute deadline) using the online application procedure.
- 2.3 The application must include:
 - 2.3.1 Transcript of Records with modules of at least 120 credits for a six-semester bachelor's program, at least 150 credits for a seven-semester bachelor's program, and at least 180 credits for an eight-semester bachelor's program; the Transcript of Records must be issued by the relevant examination authority or the relevant academic programs office,
 - 2.3.2 curriculum vitae formatted as a table
 - 2.3.3 an English-language written statement (max. 1-2 A4 pages) of the reasons for choosing the Master's Degree Program in Geodesy and Geoinformation at the Technical University of Munich, in which the candidate explains those specific abilities and interests that make him/her particularly qualified for the program; a candidate's exceptional motivation and commitment is to be demonstrated by providing details on program-related vocational training, internships, stays abroad, or program-related further education beyond the attendance and course requirements of the bachelor's program, if necessary by appropriate documentation,
 - 2.3.4 a declaration that the written statement is the applicant's own work and that the applicant has clearly identified any ideas taken from outside sources.

3. Aptitude Assessment Commission

- 3.1 ¹Aptitude Assessment is administered by the Aptitude Assessment Commission and the Selection Committees. ²The Commission is responsible for preparing the process, organizing it and ensuring a structured and standardized process for determining aptitude within the framework of these Regulations; it bears responsibility, insofar as no other body is specified by these Regulations or through delegation of its authority to another body. Selection Committees are to conduct the assessment process in accordance with No. 5 below, subject to No. 3.2 Sentence 11.
- 3.2 ¹The Commission consists of five members. ²Members of the Commission are appointed by the Dean, in consultation with the Vice Dean of Academic and Student Affairs, from among the authorized examiners of the TUM School of Engineering and Design, who are members of the degree program faculty. ³At least three Commission members must be university educators within the meaning of the Bavarian Act on Higher Education Staff (*BayHSchPG*). ⁴The departmental student council has the right to name a student representative to serve on the Commission in an advisory capacity. ⁵A deputy is to be appointed for each member of the Commission. ⁶The Commission elects a chairperson and a deputy chairperson from among its members. ⁷Procedures are governed by § 30 of the TUM Charter as amended. ⁸The term in office of Commission members is 1 year. ⁹Extensions of the term of office and reappointments are possible. ¹⁰Urgent decisions that cannot be postponed can be made by the chairperson on behalf of the Commission; He/She must inform the Commission of such decisions without delay. ¹¹The Academic Programs Office supports the Commission and the Selection Committee; the Commission may delegate to the Office the task of assessing formal admissions requirements in accordance with No. 4.1, as well as the determination of points to be awarded based on defined criteria for which there is no freedom of discretion involved. This includes, in particular, the conversion of grades and the calculation of the overall points earned by the applicant. The Office may also be involved in choosing the members of the Selection Committee from among the commissioners and assigning them to applicants.
- 3.3 ¹Each Selection Committee consists of two members of the TUM School of Engineering and Design, who are authorized to conduct examinations in the degree program according to Art. 62(1) Sentence 1 of the Bavarian Higher Education Act [*BayHSchG*] in conjunction with the act governing examiners at institutions of higher education [*Hochschulprüferverordnung*]. ²At least one member must be a university educator within the meaning of the Bavarian Act on Higher Education Staff (*BayHSchPG*). ³It is permissible to serve concurrently on both the Aptitude Assessment Commission and the Selection Committee. ⁴Members of the Committee are appointed by the Commission for a term of 1 year; No. 3.2 Sentence 9 applies accordingly. ⁵Different Selection Committees may be assigned to individual criteria and stages of the assessment process.

4. Admission to the Aptitude Assessment Process

- 4.1 Admission to the aptitude assessment process requires that all documentation specified in No. 2.2 has been submitted in a timely and complete fashion.
- 4.2 ¹Applicants who have fulfilled the requirements according to No. 4.1 will be assessed according to No. 5. ²Applicants not suited the program will receive a letter of rejection stating the grounds for rejection and informing them of legal remedies.

5. Stages of the Aptitude Assessment Process

5.1 First Stage:

- 5.1.1 ¹It will be assessed, on the basis of the written application documents required under No. 2.3, whether an applicant is suitable for the program according to No. 1 (First stage of the aptitude assessment process). ²The candidate's application documents will be evaluated on a scale ranging from 0 to 100 points, 0 being the worst and 100 the best possible result:

The following criteria will be applied to the evaluation:

a) **Discipline-Specific Skills and Qualifications**

¹The curricular analysis is conducted on the basis of competencies, rather than a schematic comparison of modules. ²The analysis is based on the fundamental subject groups listed in the following table of the Bachelor's Degree Program in Geodesy and Geoinformation at the Technical University of Munich.

Academic Subject Area	Credits TUM
Scientific Fundamentals (Mathematics and Geometry, Physics, Informatics)	35
Geodetic Fundamentals (Surveying, Adjustment Theory, Reference Systems, Fundamentals of Physical Geodesy, Fundamentals of Planning, Law, Visualization)	40
Discipline-Specific Subjects (Sensors and Methods, Geoinformatics, Land Management, Image Processing, Photogrammetry and Remote Sensing, Cartography, Satellite Geodesy, Physical Geodesy)	45

³If it is established that there are no significant differences in the competencies acquired (learning outcomes), a maximum of 60 points will be awarded. ⁴If this value is not a whole number, it will be rounded up. ⁵For any competencies missing from the student's undergraduate curriculum, points equivalent to the amount of module credits for the respective competencies in the TUM Bachelor's Program in Geodesy and Geoinformation will be deducted from the overall score.

b) Overall Grade Average

¹The applicant will be awarded 1.5 points for each tenth that the average calculated from examinations in the amount of 120 credits is better than 3.0. ²The maximum number of points is 30. ³Negative points will not be awarded. ⁴Grades of international degrees will be converted by applying the Bavarian formula.

⁵If the candidate has submitted a degree certificate containing more than 120 credits with the application, the assessment will be made on the basis of the best graded modules in the amount of 120 credits. ⁶The applicant needs to submit a list of the results together with the application and confirm its accuracy in writing.

⁷If the candidate submits this list, the average is calculated from graded module examinations with the best grades amounting to 120 credits; if no list is submitted, the overall average of grades submitted by the candidate will be used to calculate the average.

⁸The overall grade average is calculated as a weighted grade average. ⁹The grade weights of the individual modules correspond to the credits assigned to each module. ¹⁰In the process of determining grades, only the first digit after the decimal point is taken into account. All other digits are dropped without rounding.

c) Letter of Motivation

¹The applicant's written statement will be evaluated and graded on a scale of 0 – 10 points.

²The content of the written statement will be assessed using the following criteria:

1. description of existing expertise and specific aptitude in the field of geodesy,
2. a well-structured presentation of the connection between personal interests and degree program content,
3. a candidate's exceptional motivation and commitment is demonstrated by providing details on program-related, extracurricular activities and subject-specific supplementary skills (see 2.3.3),
4. presentation, linguistic expression, logical structure, clarity of statement.

³Committee members independently assess each of the four criteria, with the first three criteria weighted three times and the fourth criterion weighted once. ⁴The points total is calculated as the arithmetic mean of the individual assessments, rounded up to the nearest full point.

5.1.2 The points total in the first stage will be calculated as the sum of the individual evaluations, with decimal places rounded up.

5.1.3 ¹Applicants with at least 70 points will be deemed suitable. ²In those cases where it is determined that only some subject-specific requirements for the master's program are missing from undergraduate studies, the Commission may require that applicants complete fundamentals exams from the Bachelor's Degree Program in Geodesy and Geoinformation amounting to a maximum of 30 credits. ³These fundamentals exams must be successfully completed in the first year of study. ⁴Failed fundamentals exams may be repeated only once and at the next examination date.

5.1.4 Applicants who have achieved less than 50 points fail the aptitude assessment.

5.2 Second Stage:

5.2.1 ¹The remaining applicants will be invited for an aptitude assessment interview. ²During the second stage of aptitude assessment, both skills acquired during the applicant's bachelor's studies and the result of the assessment interview will be assessed. ³Interview appointments will be announced at least one week in advance. ⁴Time slots for interviews must be scheduled before expiration of the application deadline. ⁵The interview appointment must be kept by the applicant. ⁶If the applicant is unable to attend an aptitude assessment interview due to reasons beyond his/her control, a later appointment may be scheduled upon a student's well-grounded request, but no later than two weeks before the beginning of classes. ⁷Conducting the aptitude assessment interview via video conference is possible upon a student's well-founded request. ⁸If the video or audio transmission is disrupted, the interview can be continued after the disruption has been resolved or a follow-up appointment can be scheduled. ⁹In the event of repeated disruption, the aptitude assessment interview may be scheduled as a face-to-face meeting in exception to Sentence 7. ¹⁰Sentences 8 and 9 do not apply if it can be proven that the applicant is responsible for the disruption. ¹¹In this case, the aptitude assessment interview will be assessed.

5.2.2 ¹The aptitude assessment interview is to be held individually for each applicant. ²The interview will be held in English and last at least 20 but not more than 30 minutes for each applicant.

³The interview will focus on the following topics:

1. The applicant's clear understanding of the range of topics and the requirements of the master's program, as well as the extent to which the program's content corresponds to his or her own objectives in studying.
2. Questions regarding the aptitude parameters according to No. 1.3.
3. The applicant's competence in the analysis of geodetic questions and contextual relationships based on his or her approach to solving an exemplary problem.
4. Personal impression (after the interview): Applicants will be evaluated, for example, on their ability to convincingly demonstrate information using arguments and meaningful examples and appropriately respond to interview questions.

⁴The above topics may cover the documentation submitted according to 2.3. ⁵Any subject-specific academic knowledge that is to be taught in the Master's Degree Program in Geodesy and Geoinformation will not affect the decision. ⁶With the applicant's approval, a representative of the student body may sit in on the interview.

5.2.3 ¹Commission members independently assess each of the four points, for each of which the following maximum points may be awarded:

1. 20
2. 20
3. 30
4. 20

²Each member will assign points for each of the five interview topics on a scale from 0 to 90, 0 being the worst and 90 being the best possible result. ³The points total will be calculated as the arithmetic mean of the individual evaluations. ⁴Non-vanishing decimal places must be rounded up.

5.2.4 ¹The total number of points awarded in Stage 2 is the sum of the total of from No. 5.2.3 and the points from No. 5.1.1a (subject-specific qualification) and 5.1.1b (overall grade average). ²Applicants with 110 or more points will be deemed suitable. ³Applicants with an overall score of less than 110 points have failed the aptitude assessment.

5.3 Determination and Notification of Results

¹Applicants will be informed of the results of aptitude assessment through official notification. ²Applicants not suited for the program will receive a letter of rejection stating the grounds for rejection and informing them of legal remedies.

5.4 Candidate's suitability for the program, once determined in aptitude assessment, applies to all subsequent applications for this program.

6. Documentation

¹The aptitude assessment process must be documented, in particular the names of the participating members of the Selection Committee, the evaluation of the first and second stages, as well as the overall results. ²The aptitude assessment interview must be documented, including the date, duration and location of the assessment, the names of the participating Selection Committee members, the applicant's name, and a list of main topics of discussion in bullet points.

7. Repeat Examinations

Applicants who have failed an aptitude assessment may apply once to repeat the aptitude assessment process.

Executed following a resolution of the Senate of the Technical University of Munich dated 8 December 2021 and approval of the President of the Technical University of Munich on 14 February 2022.

Munich, 14 February 2022

Technical University of Munich

Thomas F. Hofmann
President

These Regulations were made available for inspection at the Technical University of Munich on 14 February 2022, following their announcement on 14 February 2022. Day of proclamation shall therefore be 14 February 2022.